

Aanidhya Patidar

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SUMMARY

Software Developer with a focus on applied ML and full-stack systems—trained PyTorch models, LLM-backed FastAPI services, and production-grade web applications. Built end-to-end products across AgriTech, computer vision, and smart building control, with consistent attention to measurable evaluation and reliable deployments.

EDUCATION

Vellore Institute of Technology

Bachelor of Technology in Computer Science (AI & ML)
GPA: 8.69/10.0

Bhopal, India
2023 – 2027

TECHNICAL SKILLS

Languages: Python, C++ , JavaScript

Frameworks & Libraries: FastAPI, Flask, React, Vite, PyTorch, Pandas, NumPy, Scikit-learn, LightGBM, TanStack Query, i18next

Developer Tools: Git, GitHub, VS Code, Postman, Alembic, Docker, Vercel, Render, WebSockets, REST APIs

PROJECTS

Krishi Khata — Digital Farm Ledger & Market Intelligence

[Website](#) | [GitHub](#) Aug 2025 – May 2026

Full-Stack Developer | FastAPI, React, Vite, PostgreSQL, Gemini AI, WebSockets

- Built a full-stack AgriTech PWA, orchestrating a 12-hour background worker pipeline to ingest data.gov.in commodity metrics into PostgreSQL, layered with a 1-hour in-memory TTLCache to guarantee **sub-200ms** market lookups.
- Integrated Google Gemini for runtime bilingual AI advisory (Hindi/English), engineering custom CropDataCache and TranslationCache pipelines to serve localized data locally, bypassing **redundant LLM API calls** and minimizing latency.
- Architected a secure WebSocket chat gateway with JWT authentication and enforced strict, endpoint-specific SlowAPI rate limiting (**5–20 req/min**) to successfully mitigate abuse across Auth and AI inference routes.

PixelShamer — Image Quality & Defect Detection

[Website](#) | [GitHub](#) Jan 2026 – Aug 2026

ML & Backend Engineer | IIITH Internship Assessment | PyTorch, FastAPI, React, PostgreSQL, Docker

- Built a hybrid pipeline combining a Multi-Head MLP (22 hand-crafted CV signal metrics) with a Convolutional Autoencoder for unsupervised defect localization, reaching **94.8% macro ROC-AUC** across 6 degradation classes on 150 unseen images at **<40ms** CPU inference.
- Implemented power-exponential defect gating and a 101-point PCHIP monotonic calibrator to eliminate score clustering and cut false positives by **38.8%**, without degrading classification accuracy.
- Deployed a FastAPI service with PostgreSQL persistence and Docker containerization alongside a three-viewpoint React inspection workbench, with a **10/10 passing** automated test suite.

ISOTHERM — Agentic AI Smart Building Optimization

[GitHub](#) Jul 2026

AI Developer | Honeywell Campus Connect Hackathon | Python, Agentic AI, HVAC Control

- Built a Llama 3.1-powered agentic AI to autonomously optimize HVAC controls, boosting peak comfort by 2.9%.
- Implemented predictive thermal modeling, eliminating 100% of summer reheat waste and shedding peak loads.

CERTIFICATIONS

[Applied Machine Learning in Python](#) — University of Michigan (Coursera)

Dec 2024

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional

Mar 2025

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate

Feb 2025